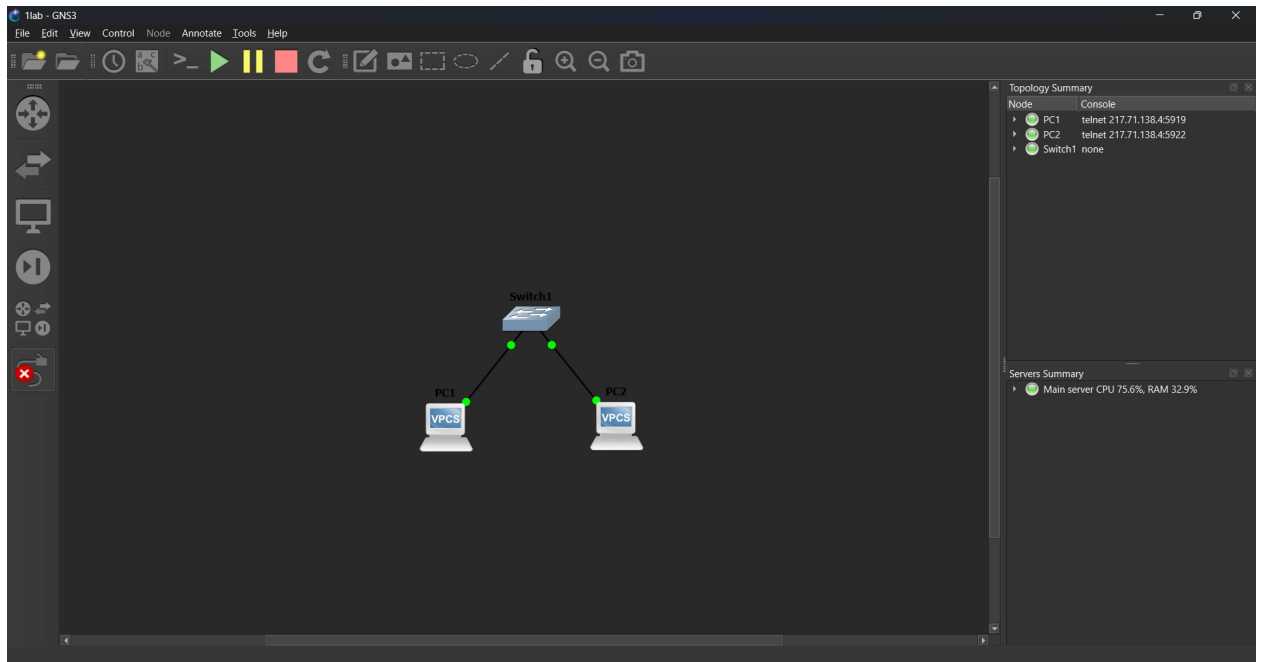




PC1

ВВОДИМ В КОНСОЛЬ:

ip 192.168.1.10/24

show ip

```
PC1 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.1.10/24
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> show ip

NAME       : PC1[1]
IP/MASK     : 192.168.1.10/24
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 23962
RHOST:PORT : 127.0.0.1:23963
MTU        : 1500

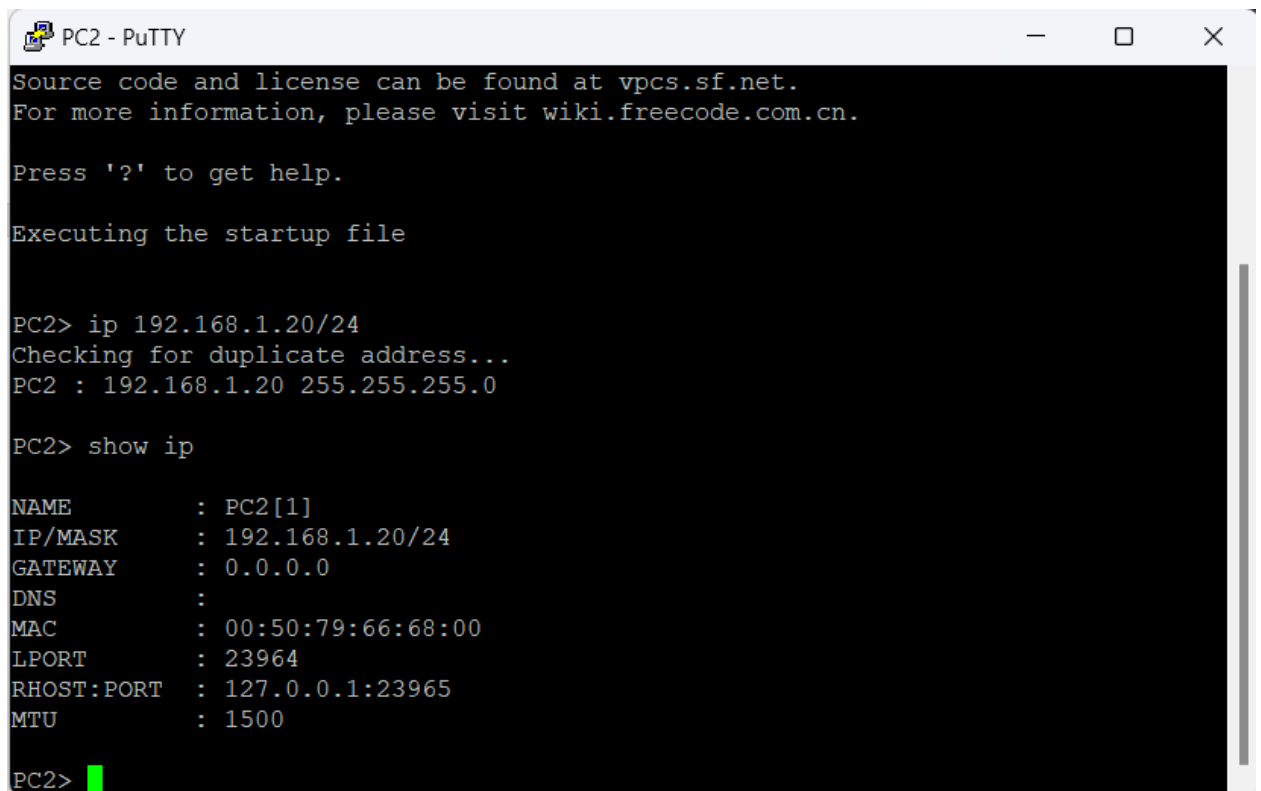
PC1>
```

PC2

ВВОДИМ В КОНСОЛЬ:

ip 192.168.1.20/24

show ip



```
PC2 - PuTTY
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.1.20/24
Checking for duplicate address...
PC2 : 192.168.1.20 255.255.255.0

PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.168.1.20/24
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 23964
RHOST:PORT : 127.0.0.1:23965
MTU        : 1500

PC2> 
```

PC1

ВВОДИМ В КОНСОЛЬ:

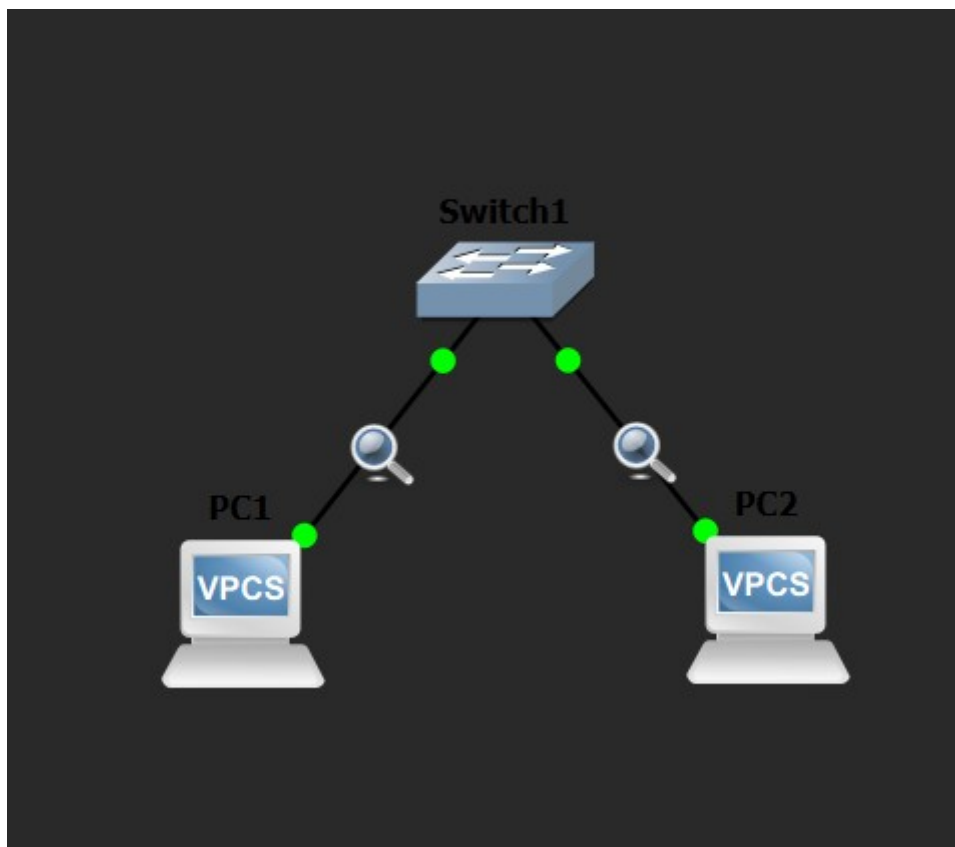
ping 192.168.1.20

```
PC1 - PuTTY
PC1> ip 192.168.1.10/24
Checking for duplicate address...
PC1 : 192.168.1.10 255.255.255.0

PC1> show ip
NAME       : PC1[1]
IP/MASK    : 192.168.1.10/24
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 23962
RHOST:PORT : 127.0.0.1:23963
MTU        : 1500

PC1> ping 192.168.1.20
84 bytes from 192.168.1.20 icmp_seq=1 ttl=64 time=0.314 ms
84 bytes from 192.168.1.20 icmp_seq=2 ttl=64 time=1.791 ms
84 bytes from 192.168.1.20 icmp_seq=3 ttl=64 time=0.284 ms
84 bytes from 192.168.1.20 icmp_seq=4 ttl=64 time=0.331 ms
84 bytes from 192.168.1.20 icmp_seq=5 ttl=64 time=0.561 ms

PC1>
```



PC1

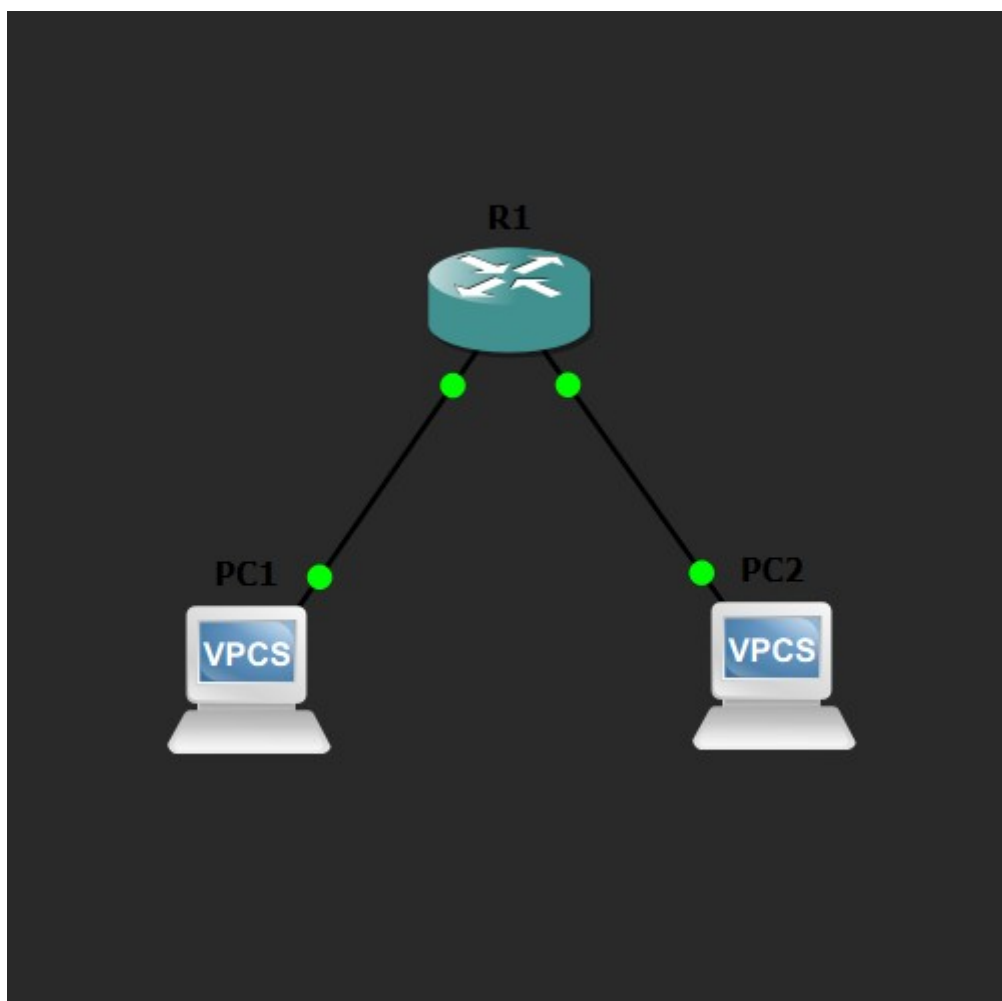
Вводим в консоль:

ping 192.168.1.20

Захват с Standard input [Switch1 Ethernet1 to PC2 Ethernet0]									
arp									
No.	Time	Delta	Source	Destination	Protocol	Length	Info		
1	0.000000		Private_66:68:01	Broadcast	ARP	64	Who has 192		
2	0.000224	0.000224	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.20		

Захват с Standard input [Switch1 Ethernet0 to PC1 Ethernet0]									
arp									
No.	Time	Delta	Source	Destination	Protocol	Length	Info		
1	0.000000		Private_66:68:01	Broadcast	ARP	64	Who has 192		
2	0.000263	0.000263	Private_66:68:00	Private_66:68:01	ARP	64	192.168.1.20		

Компьютер PC1 отправляет широковещательный ARP-запрос для определения MAC-адреса компьютера PC2. Компьютер PC2 отправляет ответ на ARP-запрос и сообщает свой MAC-адрес компьютеру PC1.



В консоль R1:

enable

configure terminal

interface FastEthernet0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface FastEthernet1/0

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

end

В консоль PC1:

ip 192.168.1.2/24 192.168.1.1

show ip

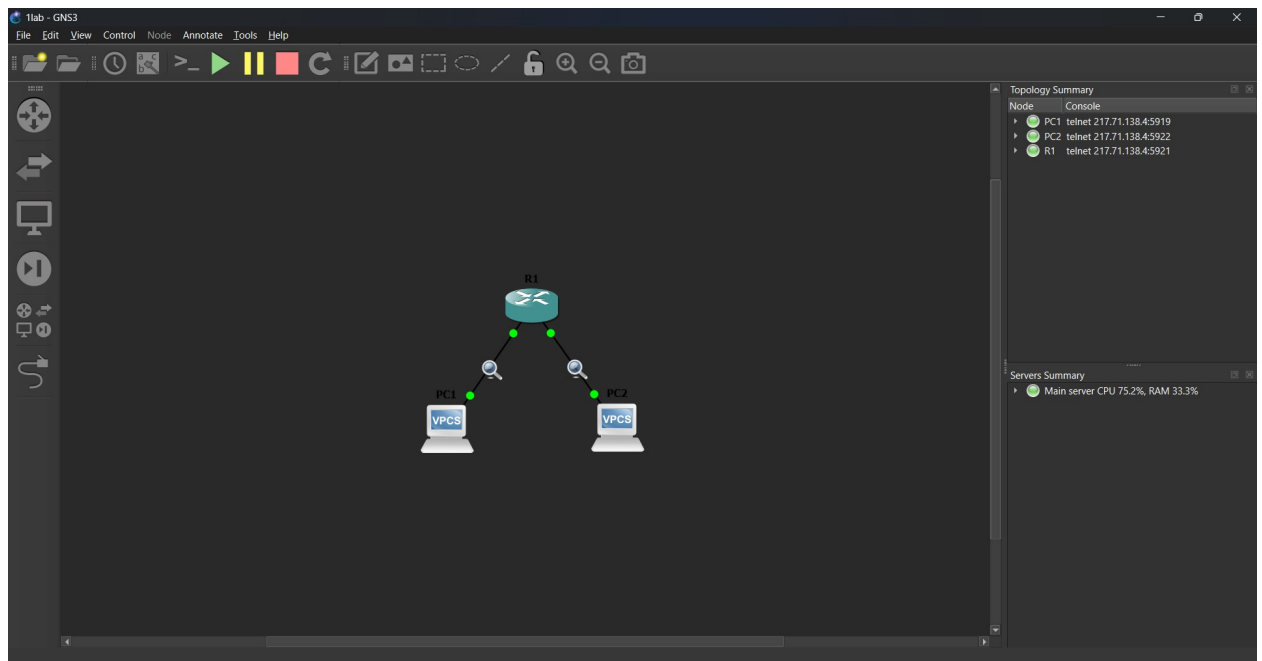
В консоль PC2:

ip 192.168.2.2/24 192.168.2.1

show ip

После проверяем на PC1:

ping 192.168.2.2



После проверяем на PC1:
ping 192.168.2.2

Захват с Standard input (R1 FastEthernet0/0 to PC1 Ethernet0)									
No.	Time	Delta	Source	Destination	Protocol	Length	Info		
15	121.081578		Private_66:68:01	Broadcast	ARP	64	Who has 192.168.2.2		
16	121.089238	0.007660	cc:01:b7:34:00:10	Private_66:68:01	ARP	60	192.168.1.1		
17	121.090868	0.001630	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
18	121.120194	0.029286	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
19	122.122315	1.002161	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
20	122.135201	0.012886	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
21	123.137232	1.002031	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
22	123.148562	0.011330	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
23	124.149619	1.001057	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
24	124.162919	0.013299	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
25	125.164418	1.001500	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
26	125.176983	0.012565	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		

<pre> > Frame 15: Packet, 64 bytes on wire (512 bits), 6 > Ethernet II, Src: Private 66:68:01 (00:50:79:66: 0020 ff ff ff ff ff c0 a8 01 01 00 00 00 00 0030 00 00 00 00 00 00 00 00 00 00 00 00 00 </pre>									
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No.	Time	Delta	Source	Destination	Protocol	Length	Info		
15	121.099418		192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
16	121.099654	0.000236	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.2.2		
17	121.109748	0.010094	cc:01:b7:34:00:10	Private_66:68:01	ARP	60	192.168.2.1		
18	121.110986	0.001238	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
19	122.124669	1.013683	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
20	122.124858	0.000189	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
21	123.138387	1.013529	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
22	123.138669	0.000282	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
23	124.152570	1.013901	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
24	124.153123	0.000553	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		
25	125.166985	1.013862	192.168.1.2	192.168.2.2	ICMP	98	Echo (ping)		
26	125.168447	0.001462	192.168.2.2	192.168.1.2	ICMP	98	Echo (ping)		

<pre> > Frame 15: Packet, 98 bytes on wire (784 bits), 9 > Ethernet II, Src: cc:01:b7:34:00:10 (cc:01:b7:34 0020 02 02 08 00 ff 77 20 93 00 01 08 09 0a 0b 0c 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 0060 3e 3f </pre>									
---	--	--	--	--	--	--	--	--	--

При выполнении команды ping между компьютерами, находящимися в разных подсетях, сначала используется протокол ARP для определения MAC-адреса шлюза по умолчанию. После получения MAC-адреса маршрутизатора начинается обмен ICMP-пакетами (Echo Request и Echo Reply), что подтверждает корректную работу маршрутизации и доступность удаленного узла.